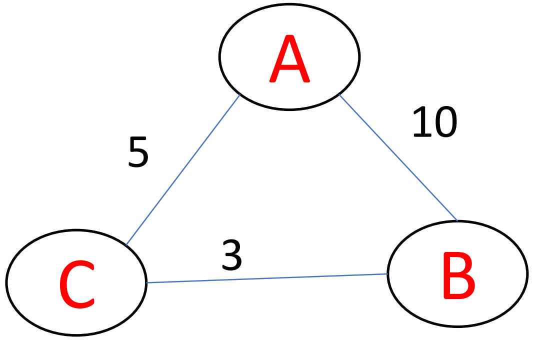
Introduction spanning Trees

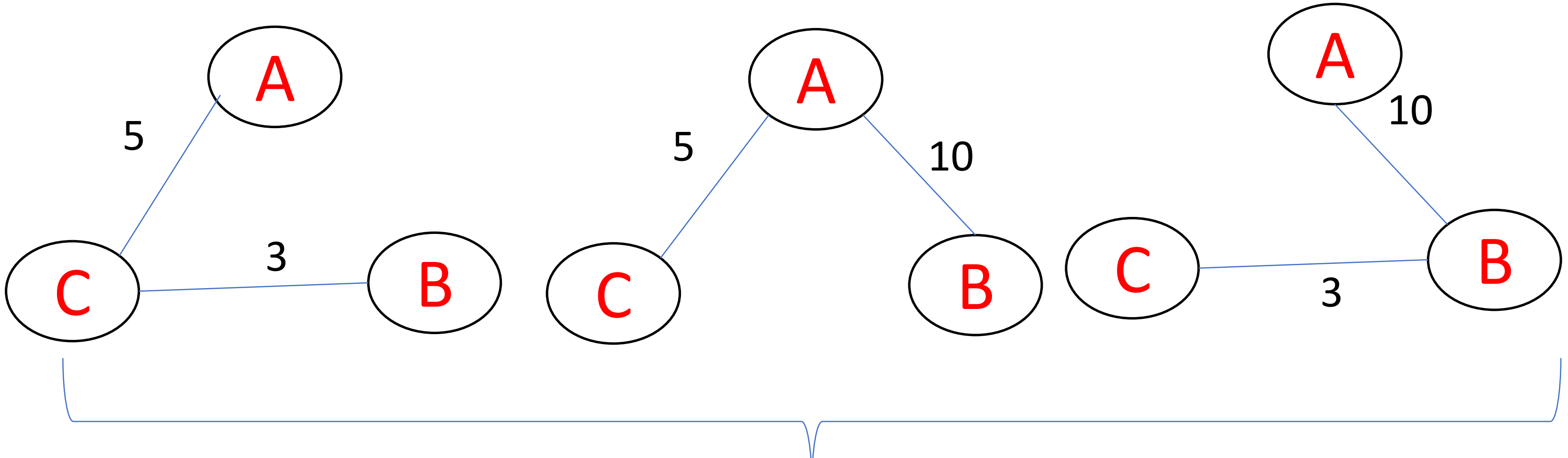
Introduction to Spanning Trees

- A spanning tree is a subset of graph G, which has all the vertices covered with minimum possible number of edges.
- A spanning does not have cycles as well as every vertex is connected.
- Hence, every connected undirected graph has at least one spanning tree.

No. of spanning trees for a complete undirected graphs $n^{(n-2)}$ where n is number of vertices.



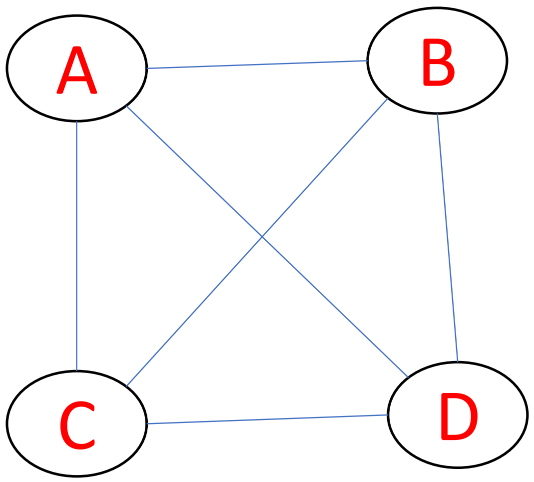
Graph G



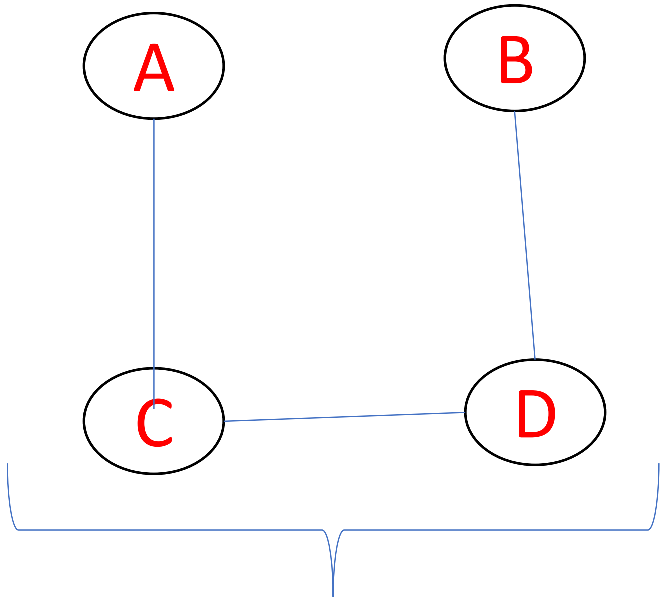
Spanning Trees

Examples

No. of edges in a spanning tree are n-1, where n is number of vertices.

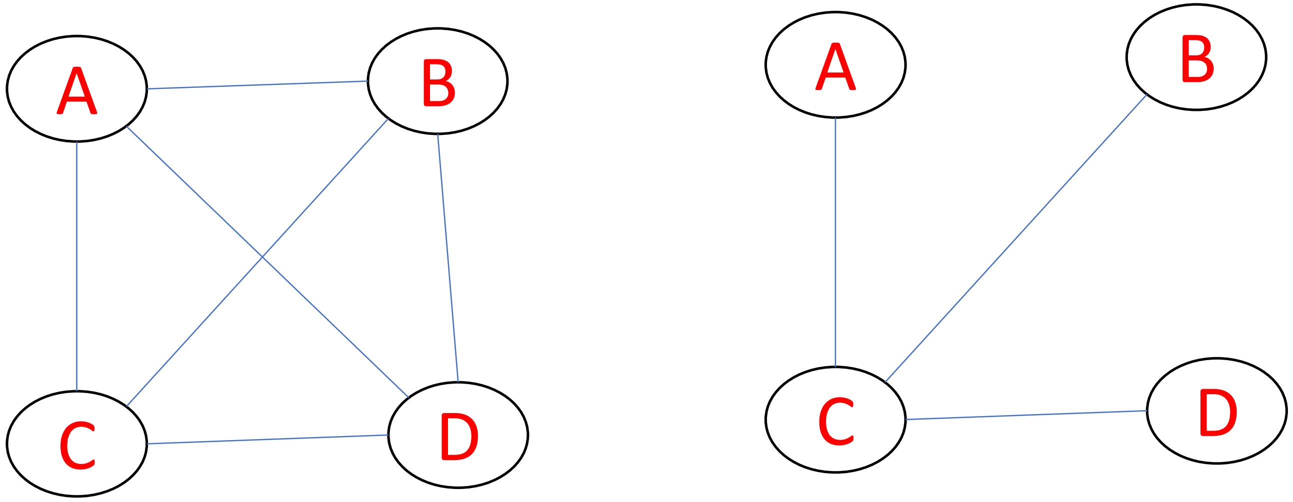


Graph G



Spanning Trees

Examples



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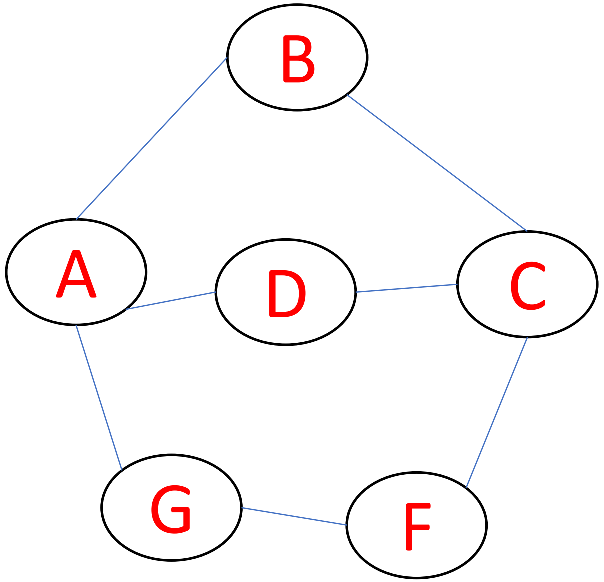
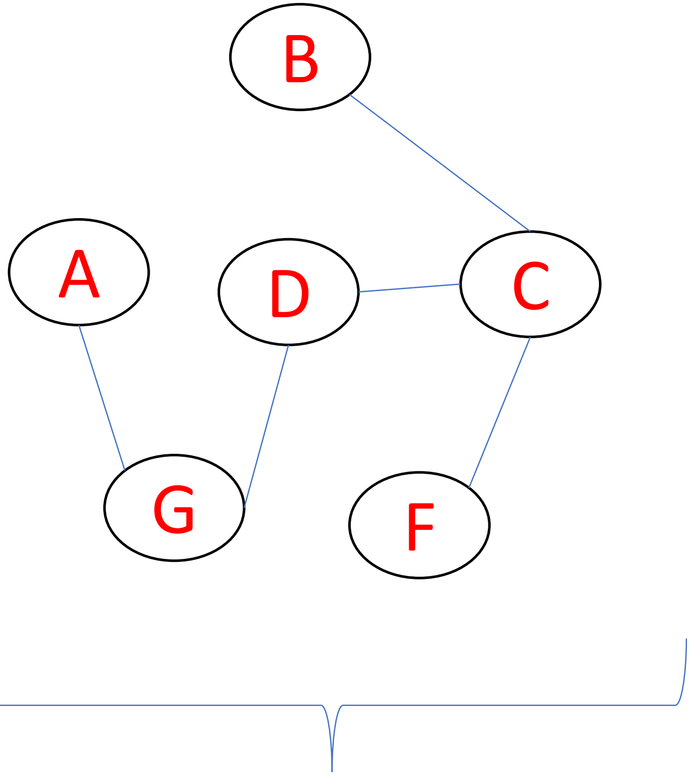


Graph G



Spanning Trees

Properties of spanning trees

- A connected graph G can have more than one spanning tree.
- A disconnected graph do not have spanning trees.
- All possible spanning trees of graph G have the same number of (n-1) edges and vertices.
- A spanning tree does not have any cycle (loops).
- Removing one edge from the spanning tree will make the graph disconnected, it means the spanning tree is minimally connected.
- Adding one edge to the spanning tree will create a circuit, it means the spanning tree is maximally acyclic.

Mathematical Properties of spanning trees

- Spanning tree has n-1 edges, where n is the number of nodes (vertices).
- A complete graph can have maximum n^n-2 number of spanning trees.

Applications of spanning trees

- Spanning tree is basically used to find a minimum path to connect all nodes in a graph. Common applications of spanning trees are civil network planning, computer network routing protocol, cluster analysis etc.

Minimum spanning trees

- In a weighted graph, a minimum spanning tree is a spanning tree that has minimum weight than all other spanning trees of the same graph. In real-world situations, this weight can be measured as distance, congestion, traffic load or any arbitrary value denoted to the edges.

Minimum spanning tree Algorithm

- Two algorithms for spanning trees are krushkal's Algorithm and Prim's Algorithm.